INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A.**, Semester: **Third**

Stream: **B.C.A.**

Paper Code: **BCACC303**

Paper Name: **Python Programming**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. Name any two functions used for type conversion in Python. **2**

2. List any four Python keywords. **2**

3. Write a recursive function definition for factorial. **2**

4. Differentiate between =and ==operators in Python. **2**

5. What is the difference between function definition and function call? **2**

6. What is the output of the following code? **2**

```
print("Python" * 3)
```

7. Does the **break** statement exit only from the innermost loop? **2**

8. What is the base condition in recursive functions? **2**

9. Write a Python statement to delete an element from a list. **2**

10. Write the output of: **2**

```
print(max("apple"))
print(min("apple"))
```

11. How do you access the first and last character of a string in Python? **2**

12. Write a code snippet to demonstrate multiple inheritance. **2**

13.	Name any two built-in class attributes in Python.	**2**
14.	What is method overriding?	**2**
15.	What is the purpose of abstraction in Python OOP?	**2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16.	Write a Python program to accept radius and compute area and circumference of a circle.	**5**
17.	Write a Python function to find the factorial of a number.	**5**
18.	Explain the role of nested conditions in Python with an example.	**5**
19.	Write a program to check whether a given year is a leap year or not.	**5**
20.	Write a Python function to check whether a number is prime or not.	**5**
21.	Explain the use of find(), replace(), and split()string methods with examples.	**5**
22.	Explain the process of creating and importing a user-defined module with an example.	**5**
23.	Differentiate between relational and logical operators in Python with examples.	**5**
24.	Explain string operations in Python with examples of concatenation, repetition, and slicing.	**5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25.	Write a program to count the number of uppercase, lowercase, digits, and special characters in a string.	**10**
26.	Explain function definition, parameters, arguments, and return values with suitable examples.	**10**
27.	Write a Python program to demonstrate use of tuple packing and unpacking.	**10**
28.	Discuss abstraction in OOP. Write a program using abc module to demonstrate abstract classes.	**10**
29.	Write a Python program that uses recursion to calculate the greatest common divisor (GCD) of two numbers.	**10**

30. Discuss in detail the differences between inheritance, abstraction, and polymorphism with example programs. **10**

31. Explain nested conditions and nested loops with example programs. **10**

32. Discuss with examples: similarities and differences between lists, dictionaries, and tuples. **10**

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